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Leading Ideas in the Nineteenth Century
(*Ledende Tanker i det nittende Aarhundrede*)

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Extract: pp. 28–32. The text begins and ends mid-sentence.

Translated from the Danish

[…] into one with the relation between ground and consequent, will not be discussed *[28]*
here. But for a historical consideration it is of great interest to see that the discovery of the conservation of energy was a continuation and confirmation by experience of a principle that had already been put forward in the seventeenth century. Descartes put forward a law of the conservation of motion, according to which no motion came to nothing and no motion arose from nothing; there was always the same sum of motion in the universe, only it was distributed in different ways. Leibniz criticized this doctrine, maintaining that according to it every transition from motion to rest and from rest to motion within the universe would be incomprehensible. There is, he further maintains, no absolute opposition between rest and motion; rest is arrested motion, due to an equilibrium between forces (tendencies), and motion is released anew as soon as the equilibrium is removed. It is thus not motion but its possibility, "force", that is conserved, and while there is not always the same sum of motion in the universe, there is the same quantity of "force" (energy, in the modern expression). Leibniz laid such great weight on this principle that he even called it the foundation of all the laws of nature (*fundamentum naturæ legum*). In the science of nature of the following period, this principle receded for various reasons, although it has again and again been at work as a tacit presupposition. When it now came forward again, it was not as a proposition of pure mechanics, but as a principle asserting a definite ratio of conversion between the various special forces of nature (motion, heat, light, electricity, etc.), and it was now carried through by experimental investigations. It now stands as a law for all material manifestations of force. — Nowhere is it so clear as here that the science of the nineteenth century, in all its greatness, consists after all in carrying through ideas that stem from the remarkable time when modern science was founded.

It has been said of the principle of the conservation of energy that it is only

a generalization from a limited experience. It is also true that a confirmation by experience of a principle such as this can in a certain sense never be concluded. One will never be able to isolate and close off a part of the world in such a way that every supply or removal of energy would be absolutely excluded. And there always remains the possibility that the principle holds only for the part or parts of the world that we have so far been able to investigate. And since "the world" can never come to lie before us as a whole concluded once and for all, it is an infinite task that is set
[29] for research here. But the principle of the conservation of energy here only shares the fate of all the propositions to which research into actual nature can lead. The concept "world" in the sense of absolute totality has the scientific significance that it shows us the possibility of ever new tasks as the horizon widens. And it marks a significant advance that the principle put forward in the seventeenth century has now proved accessible to progressive confirmation by experience. Like every great hypothesis, the principle supplies an important maxim for research, namely, always to look for new forms of energy as equivalents for those that disappeared, and to look for the preceding forms of energy from which the new ones have arisen. Maxwell has particularly emphasized the epistemological significance of the principle of the conservation of energy, that it leads us, at every new natural phenomenon, to put definite questions. This is indeed the significance of all living principles in science, and this significance cannot be overstated.

The principle says that in every material change there is only a conversion of energy, without any energy being lost; but it says nothing about the direction in which the conversions take place, and in particular nothing about whether a conversion in the opposite direction can always as easily take place. A French investigator had, as early as eighteen years before Mayer's treatise, put forward a principle that throws light on this matter. Sadi Carnot (*Réflexions sur la puissance motrice du feu.* 1824) showed that, for heat to be a motive force, there must be a difference of temperature. Motion is produced only when such a difference arises between the states of the bodies between which the action is to take place. But since temperature always tends toward equilibrium, heat passing over from warmer to colder bodies and thereby being distributed, the possibility of a new conversion of heat into motion thereby falls away within the same circle of bodies, considered as a closed whole.

Carnot's principle, too, holds only for a limited world. And if by "the world" one understands a limited whole, there prevails in it a process that leads in the direction of a complete equilibrium, a state in which no new release of energy is possible.

Carnot's principle, which, like Mayer's, had to wait long for full recognition, does not conflict with the principle of the conservation of energy. For in the state of equilibrium that there is a constant tendency to bring about, no energy has been lost; only, all energy has passed over from being actual to being potential.

An optimistic philosophy of nature will hold to Mayer's principle, a pessimistic *[30]* one to Carnot's. But both principles hold only for a limited horizon, and since science always seeks to widen its horizon, the discussion of which standpoint is the decisive one will always have to be taken up anew. —

A still more intimate connection between the various forces of nature than that which follows from the principle of the conservation of energy was shown when it was established that forces of nature which had so far stood as different in kind follow the same laws and must therefore be assumed to be of the same nature. Thus Melloni and other investigators showed that light and heat were one. H. C. Ørsted had already divined a connection between light and electricity, conceiving the lightning flash as a series of electric sparks. And after Maxwell had by a mathematical route put forward the electromagnetic theory of light, according to which light can be traced back to electricity, the laws of these two activities being the same, Hertz showed experimentally that light and electricity, as regards wave motion, velocity, refraction, etc., present the same relations. Optics could now be regarded as an appendix to the theory of electricity.

On the whole, the confirmation by experience that the principle of the conservation of matter and force gained in the nineteenth century is far from having brought the scientific conception of the world to a conclusion. It was in itself a magnificent picture of the world that was unrolled. The world came to stand as a whole within which constant conversions of matter and force took place, without any matter or force at all arising or perishing. As we have already seen, this picture of the world cannot be concluded, since the widening of the horizon can always bring new possibilities. But on another side too, in the very foundation, research goes beyond the picture of the world just indicated. The development of the theory of electricity has led to this: that far down beneath the matters and forces of earlier science a new and rich world comes into view, in which perhaps the real foundation of the world of matter and force in which we move is to be sought. The theory of electricity has led to the discovery of minute particles, so-called electrons, which are smaller than Dalton's atoms and of which these must be thought to be composed. The change in the scientific conception that was thereby heralded at the end of the century is described in the following

way by a Danish physicist: "In the last half of the century the theory of electricity has pressed forward so strongly at the expense of the other forces of nature that
[31] it seems about to swallow them all. One is now inclined in advance to refer not only optics but also the other forces of nature to it; electricity alone is so manifold in its forms of manifestation that it can be thought capable of comprising them all. Even the very concept of mass can be dispensed with, if the atoms are thought of as charged with electricity, since in that case every acceleration must produce induction in the surroundings, and thereby in turn a resistance to the motion. The dissolving power of the theory of electricity has extended to chemistry as well, to which physical chemistry bears witness. Even the old concept of the atom is in danger of collapsing. Thus J. J. Thomson and Zeemann have believed they could demonstrate masses a thousand times lighter than the chemists' atoms" (C. Christiansen: "Før og nu" [Then and now], first volume of *Tidsskrift for Fysik*. 1908).

There thus opens up, at the end of the century, the prospect of an entirely new physics, which goes further back than the one founded by Galilei and Newton and will derive the maxims of the old physics from still deeper-lying principles. From the world of the great masses one has begun to descend to the world of the small elements. "The time," says an American physicist, "seems not to be distant when there will be room for a new 'Principia' and for a work that will accomplish for the molecular systems what the 'Mécanique Céliste' accomplished for the solar system." (Woodward: The Unity of Physical Science. Congress of Arts and Sciences St. Louis 1904).

If it is asked what it is, then, that moves according to the laws of electricity and under special conditions appears as light, heat, etc., the natural scientist answers with the beautiful name ether. But the ether is known only through its special forms of manifestation, among which electricity is regarded as the fundamental one. The famous French physicist Langevin declares that it is impossible to give a picture, a representation, of the ether, since its properties are entirely different from those of matter and probably far simpler than these. And he thinks that the difficulty we face here perhaps hangs together with the present constitution of human consciousness: for centuries we have been used to thinking by means of the concept of matter, and are therefore unable to form a concrete representation that does not have a material content. The ether can be thought only by means of the concepts of the electromagnetic properties we ascribe to it (Langevin: Congress of Arts and Sciences. 1904).

Plato already saw that a thoroughgoing atomic theory cannot be built up by means *[32]* of the ordinary concept of matter, cannot be grounded in sensuous representations, but must make use of analogy. The result he reached by a critical thinking-through of Democritus's atomic theory, modern physics has now reached by its own route. —

For no science have greater possibilities and tasks shown themselves at the beginning of the twentieth century than for physics. But what has already been won — the progressive carrying-through and confirmation of the great ideas that three hundred years ago led to the rise of a "new science" — is a great achievement, which will be of incalculable significance for the human conception of existence, once its scope really dawns on consciousness. Both in the great and in the small, in the world of the heavenly bodies and in that of the imperceptible elements, a greater connection has shown itself than one could have divined.

b. *The Theory of Evolution.*

The science of organic life could not remain untouched by the great movement stirring in the physical and chemical domain. Ever since the first rise of modern natural science, one had sought to bring the doctrine of life under the general laws of nature as physics demonstrates them. The principle that all change is motion, one sought to maintain not only in the domain of inorganic but also in that of organic nature. Descartes insisted emphatically that organisms were only mechanisms, set going according to the general laws of motion, and he held that Harvey's discovery of the circulation of the blood confirmed this view, which became the prevailing one in the seventeenth century. As a reaction against this mechanical view there appeared the so-called vitalism, which maintained that organic phenomena could find their explanation only if one assumed a peculiar life force, independent of all physical and chemical forces. Vitalism reached its most pronounced form in the medical school of Montpellier in the eighteenth century. The "life force" directs the physical and chemical processes in the organism in definite directions, brings about a connection among everything that goes on in the living being, so that it can subsist as a whole, provides for the formation of the substances that life needs, and for sufficient strength to overcome the resistance that life meets from without.

The nineteenth century appears here, as on other points, as a carrying-through and specialization of the fundamental ideas of the seventeenth century. Mechanism and vitalism at bottom designate only two [...]